

Lecture 10b

EE-215 Electronic Devices and Circuits

Asst Prof Muhammad Anis Ch

BJT: Device Structure and Physical Operation

The pnp Transistor

- figure shows a pnp transistor biased to operate in the active mode

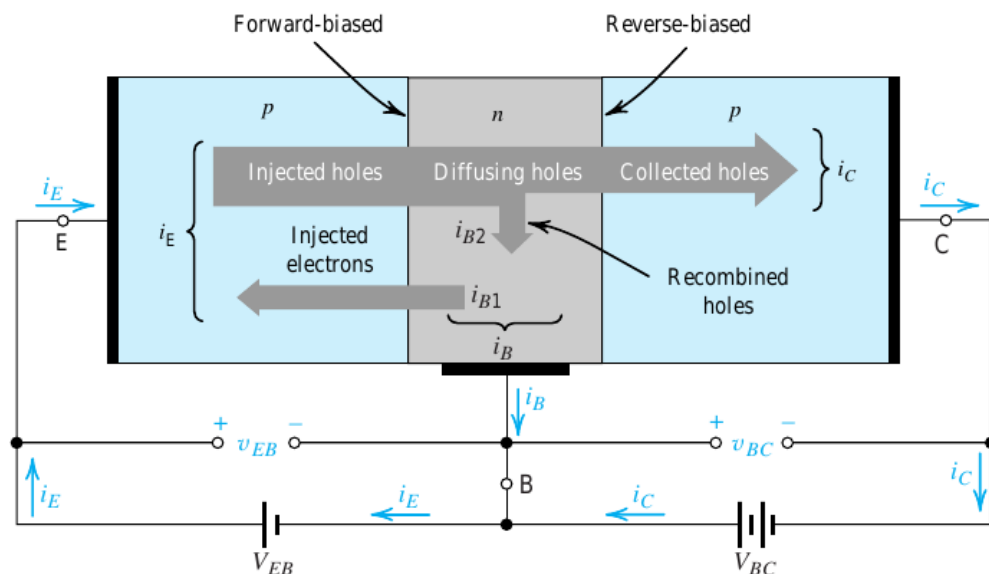
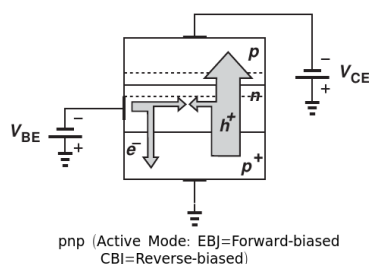


Figure 6.10 Current flow in a pnp transistor biased to operate in the active mode.

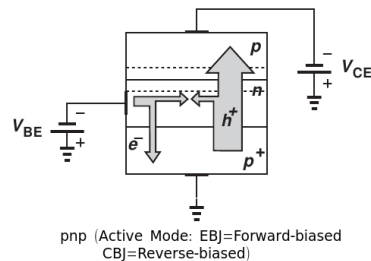
- to achieve active mode of operation, EBJ is forward-biased while CBJ is reverse-biased
 - Unlike the npn transistor, current in the pnp transistor
 - is mainly conducted by holes, injected from the emitter
 - into the base as a result of the forward-bias voltage v_{EB}



- These holes injected into the base diffuse through the
 - base region into the depletion region of CBJ and are
 - then swept away into the collector region under the
 - influence of the electric field, resulting in i_C
- A small number of base majority carriers (electrons) are
 - injected into the emitter (forming i_{B1}) or recombined with the

- holes in the base region (forming i_{B2}), thus creating the base current ($i_B = i_{B1} + i_{B2}$)
- thus the pnp transistor operates in a manner similar to that of the npn transistor

- $\Rightarrow$ the current-voltage relations for the pnp transistor are
 - identical to that of the npn transistor except that v_{BE} has to be replaced by v_{EB}



- $\Rightarrow$ for a pnp transistor

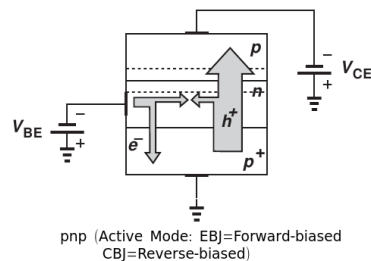
- $i_C = I_S e^{v_{EB}/V_T}$

- $i_B = \frac{i_C}{\beta} = \frac{I_S}{\beta} e^{v_{EB}/V_T}$

- $i_E = \frac{i_C}{\alpha} = \frac{I_S}{\alpha} e^{v_{EB}/V_T}$

- $i_E = i_C + i_B$

- where $\alpha = \frac{\beta}{\beta+1}$, $\beta = \frac{\alpha}{1-\alpha}$



- Also the large signal operation of a pnp device in active mode can be modeled as

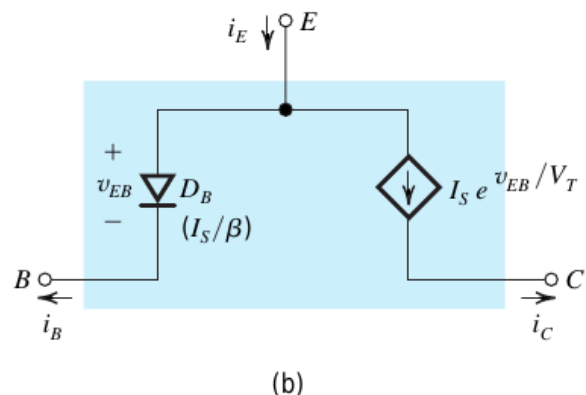
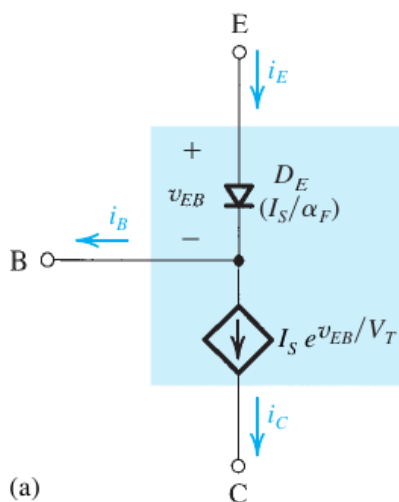
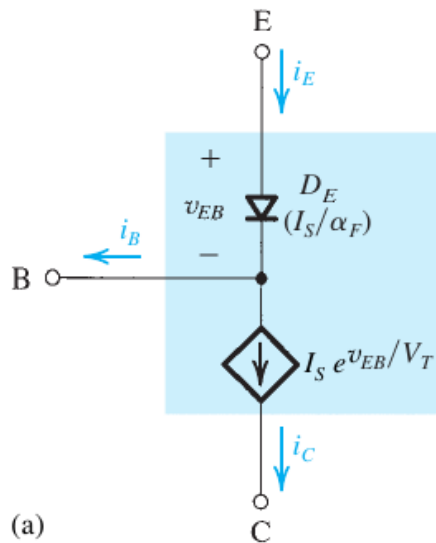
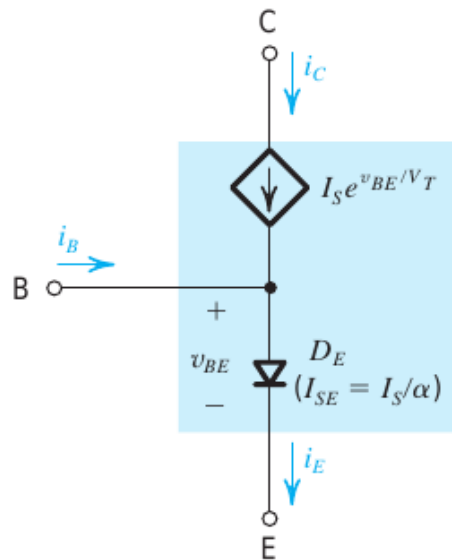


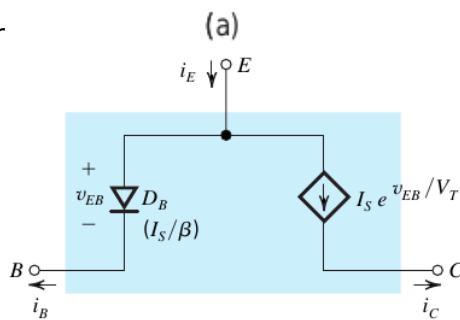
Figure 6.11 Two large-signal models for the pnp transistor operating in the active mode.



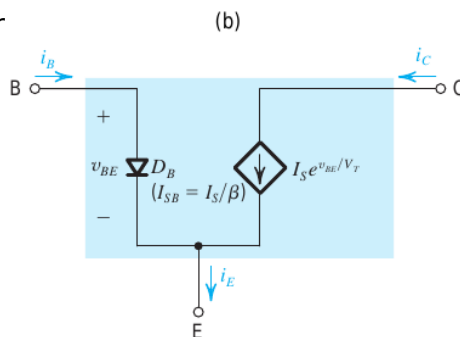
• (a)



• The npn Transistor



• The pnp Transistor



• The npn Transistor

(c)

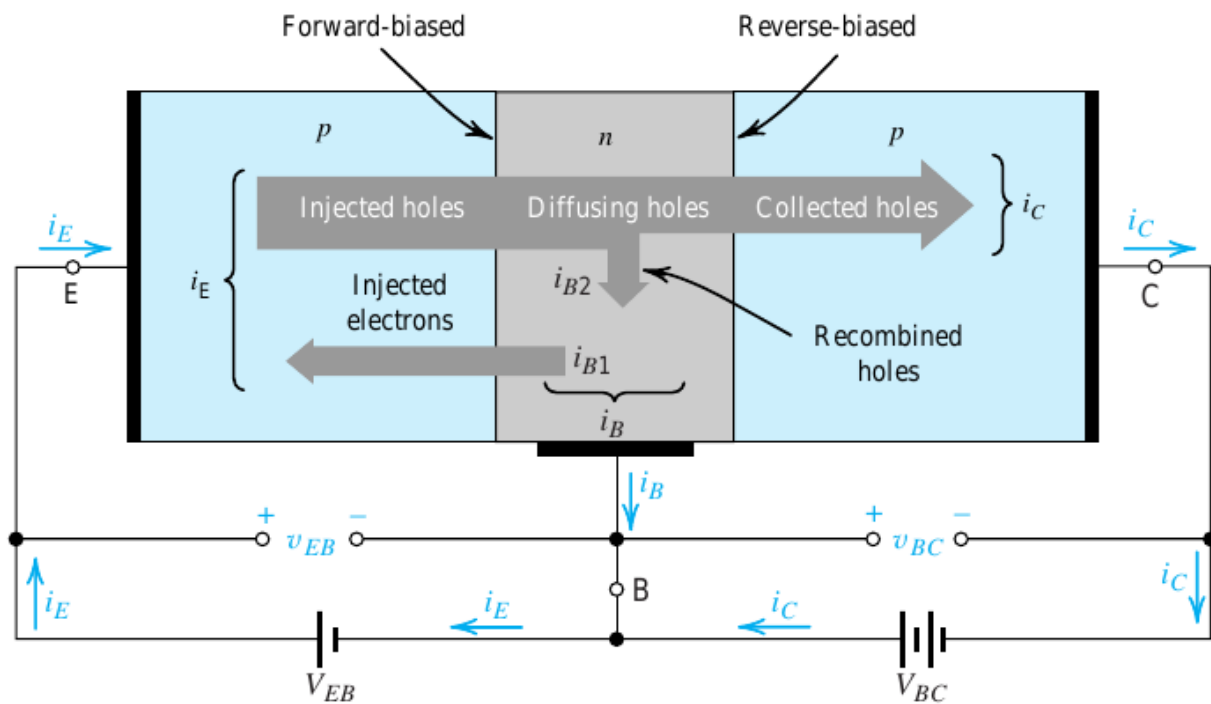


Figure 6.10 Current flow in a *pnp* transistor biased to operate in the active mode.

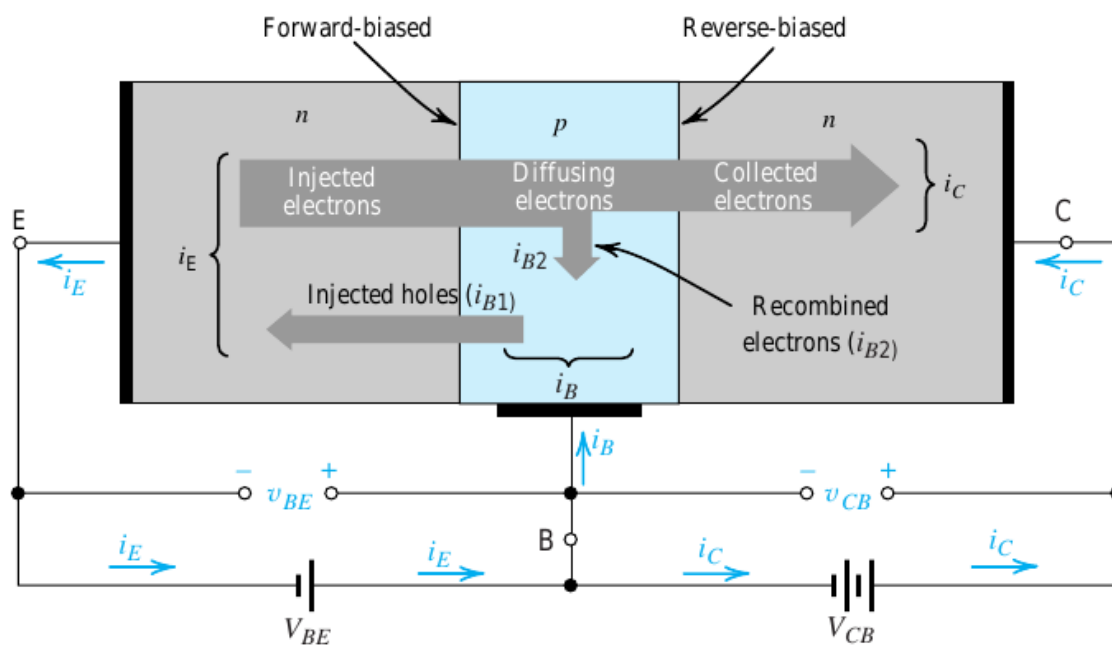
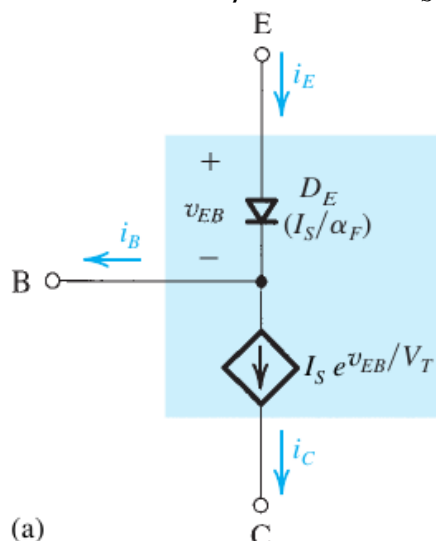


Figure 6.3 Current flow in an *nnp* transistor biased to operate in the active mode. (Reverse current components due to drift of thermally generated minority carriers are not shown.)

Exercise 6.10

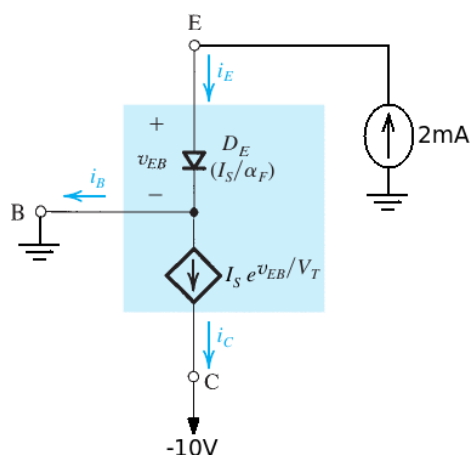
- Consider the model in Fig. 6.11(a) applied in the case of a pnp transistor whose base is grounded, the emitter is fed by a constant-current source that supplies a 2mA current into the emitter terminal, and the collector is connected to a -10V dc supply. Find the emitter voltage, the base current, and the collector current if

for this transistor $\beta = 50$ and $I_S = 10^{-14}A$.



Solution: Exercise 6.10

- here $\beta = 50$, $I_S = 10^{-14}A$, $i_E = 2mA$
 - $v_E = ?$, $i_B = ?$, $i_C = ?$
 - $\alpha = \frac{\beta}{\beta+1} = \frac{50}{51} = 0.9804$
 - $\Rightarrow i_E = 2mA = \frac{i_C}{\alpha} = \frac{i_C}{0.9804}$
 - or $i_C = 0.9804i_E = 0.9804 \times 2m = 1.9608mA$
 - $i_B = \frac{i_C}{\beta} = \frac{1.9608m}{50} = 39.216\mu A$
 - $I_S e^{v_{EB}/V_T} = i_C = 1.9608mA$
 - $\Rightarrow v_{EB} = V_T \ln \frac{1.9608m}{I_S} = 0.65004V$
 - $\Rightarrow v_E - v_B = v_E - 0 = 0.65004V$ or $v_E = 0.65004V$



Exercise 6.11

- For a pnp transistor having $I_S = 10^{-11}A$ and $\beta = 100$, calculate v_{EB} for $i_C = 1.5A$.

Solution: Exercise 6.11

- here $I_S = 10^{-11}A$, $\beta = 100$, $i_C = 1.5A$

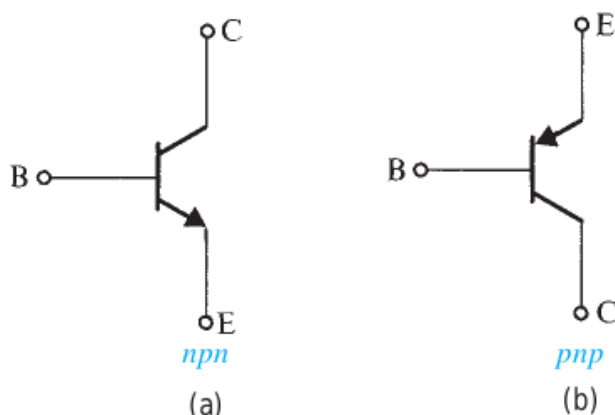
- $v_{EB} = ?$
- $i_C = I_S e^{v_{EB}/V_T}$
- $\Rightarrow 1.5 = 10^{-11} e^{v_{EB}/V_T}$
- or $e^{v_{EB}/V_T} = \frac{1.5}{10^{-11}}$
- take natural log on both sides
- $\frac{v_{EB}}{V_T} = \ln\left(\frac{1.5}{10^{-11}}\right)$
- $v_{EB} = V_T \ln\left(\frac{1.5}{10^{-11}}\right) = 25m \ln\left(\frac{1.5}{10^{-11}}\right) = 0.64335V$

BJT:

Current-Voltage Characteristics

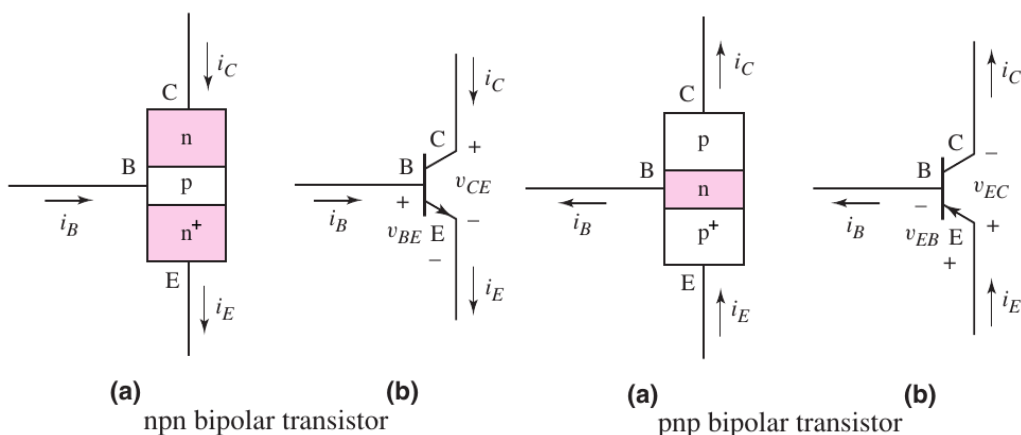
Circuit Symbols and Conventions

- the circuit symbols for an npn and pnp transistor are indicated in fig



◦ **Figure 6.12** Circuit symbols for BJTs.

- In both symbols, the emitter is distinguished by an arrowhead.
- this distinction is important as practical BJT is not a symmetrical device



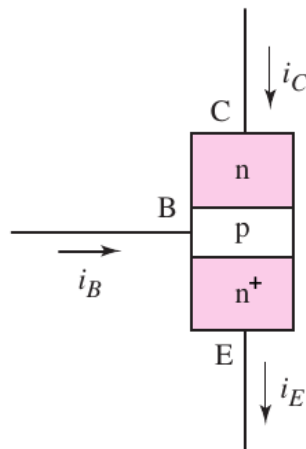
- The polarity of the device i.e. npn or pnp
 - is indicated by the direction of the arrowhead on the emitter.
 - This arrowhead points in the direction of the conventional current flow.
- Recall that in npn BJT, current flows from the collector to emitter
 - (as the electrons are flowing from emitter to the collector)
 - in a pnp BJT, the current flows from emitter to the collector

- (as the holes flow from emitter to the collector)

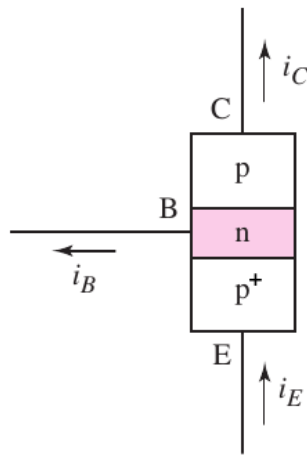
- A summary of the BJT current-voltage relationships in the active mode of operation can be given as

Table 6.2 Summary of the BJT Current-Voltage Relationships in the Active Mode	
$i_C = I_S e^{v_{BE}/V_T}$ $i_B = \frac{i_C}{\beta} = \left(\frac{I_S}{\beta}\right) e^{v_{BE}/V_T}$ $i_E = \frac{i_C}{\alpha} = \left(\frac{I_S}{\alpha}\right) e^{v_{BE}/V_T}$	
Note: For the <i>pnp</i> transistor, replace v_{BE} with v_{EB} .	
$i_C = \alpha i_E$ $i_C = \beta i_B$ $\beta = \frac{\alpha}{1 - \alpha}$	$i_B = (1 - \alpha)i_E = \frac{i_E}{\beta + 1}$ $i_E = (\beta + 1)i_B$ $\alpha = \frac{\beta}{\beta + 1}$
$V_T = \text{thermal voltage} = \frac{kT}{q} \approx 25 \text{ mV at room temperature}$	

- Note that an npn transistor, whose EBJ is forward-biased,
 - will operate in the active mode as long as
 - the collector voltage doesnot fall below that of the base by
 - more than approximately 0.4V
 - i.e. $v_{CB} > -0.4V$

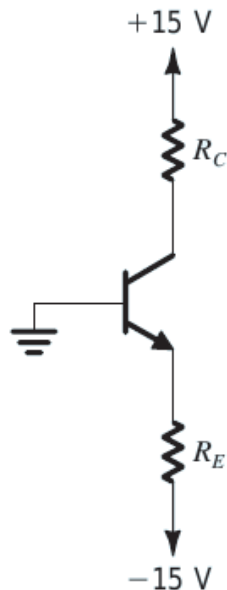


- for a pnp transistor whose EBJ is forward-biased
 - will operate in the active mode as long as
 - the collector voltage is not allowed to rise above that
 - of the base by more than 0.4V
 - i.e. $v_{CB} < 0.4V$
 - or $v_{BC} > -0.4V$



Example 6.2

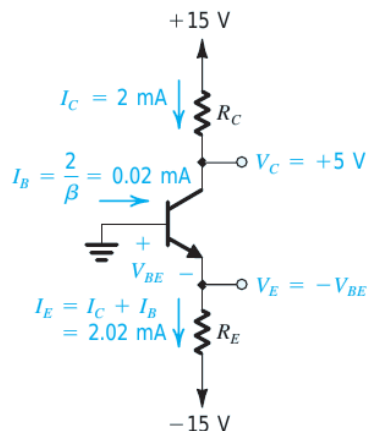
- The transistor in the circuit of Fig. 6.14(a) has $\beta = 100$ and exhibits a v_{BE} of $0.7V$ at $i_C = 1mA$. Design the circuit so that a current of $2mA$ flows through the collector and a voltage of $+5V$ appears at the collector.



Solution: Example 6.2

- here npn, $\beta = 100$, $v_{BE1} = 0.7V$ at $i_{C1} = 1mA$
 - $i_C = 2mA$, $v_C = 5V$
 - $R_C = ?$, $R_E = ?$
 - As $i_C = I_S e^{v_{BE}/V_T}$
 - $\Rightarrow i_{C1} = I_S e^{v_{BE1}/V_T}$, $i_{C2} = I_S e^{v_{BE2}/V_T}$
 - dividing
 - $\frac{i_{C2}}{i_{C1}} = \frac{I_S e^{v_{BE2}/V_T}}{I_S e^{v_{BE1}/V_T}} = e^{(v_{BE2} - v_{BE1})/V_T}$
 - take natural log on both sides
 - $\ln \frac{i_{C2}}{i_{C1}} = (v_{BE2} - v_{BE1})/V_T$
 - $v_{BE2} - v_{BE1} = V_T \ln \frac{i_{C2}}{i_{C1}}$

$$\blacksquare v_{BE2} = v_{BE1} + V_T \ln \frac{i_{C2}}{i_{C1}}$$



$$\bullet v_{BE2} = v_{BE1} + V_T \ln \frac{i_{C2}}{i_{C1}}$$

◦ here $v_{BE1} = 0.7V$ at $i_{C1} = 1mA$, $i_{C2} = i_C = 2mA$, $v_{BE2} = v_{BE} = ?$

$$\blacksquare = \Rightarrow v_{BE} = v_{BE2} = 0.7 + (25m) \ln \frac{2m}{1m} = 0.717V$$

■ As the base is grounded

$$\blacksquare = \Rightarrow v_{BE} = 0.717 = v_B - v_E = 0 - v_E$$

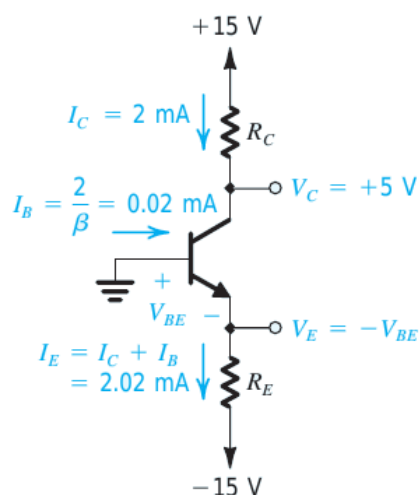
$$\blacksquare \text{ or } v_E = -0.717V$$

$$\blacksquare \alpha = \frac{\beta}{\beta+1} = \frac{100}{101} = 0.9901$$

$$\blacksquare i_E = \frac{i_C}{\alpha} = \frac{2m}{0.9901} = 2.02mA$$

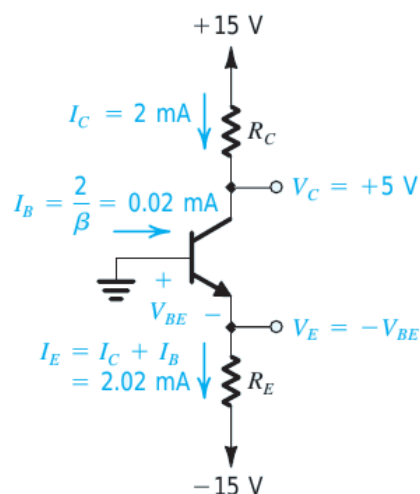
■ By ohm's law

$$\blacksquare R_C = \frac{15 - v_C}{i_C} = \frac{15 - 5}{2m} = \frac{10}{2m} = 5k\Omega$$



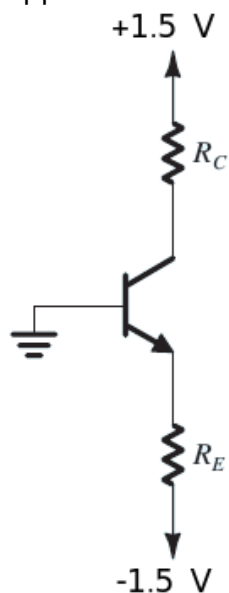
• also by ohm's law

$$\circ R_E = \frac{v_E - (-15)}{i_E} = \frac{-0.717 - (-15)}{2.02m} = \frac{-0.717 + 15}{2.02m} = 7.0708k\Omega$$



Exercise D6.12

- Repeat Example 6.2 for a transistor fabricated in a modern integrated-circuit process. Such a process yields devices that exhibit larger v_{BE} at the same i_C because they have much smaller junction areas. The dc power supplies utilized in modern IC technologies fall in the range of 1 V to 3 V . Design a circuit similar to that shown in Fig. 6.14 except that now the power supplies are $\pm 1.5\text{ V}$ and the BJT has $\beta = 100$ and exhibits v_{BE} of 0.8 V at $i_C = 1\text{ mA}$. Design the circuit so that a current of 2 mA flows through the collector and a voltage of $+0.5\text{ V}$ appears at the collector.



Solution: Exercise D6.12

- here $\beta = 100$, $v_{BE1} = 0.8\text{ V}$ at $i_{C1} = 1\text{ mA}$
 - $i_C = 2\text{ mA}$, $v_C = 0.5\text{ V}$

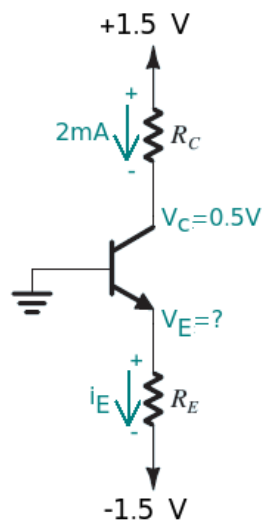
■ By ohm's law

$$R_C = \frac{1.5 - v_C}{i_C} = \frac{1.5 - 0.5}{2\text{ mA}} = 500\Omega$$

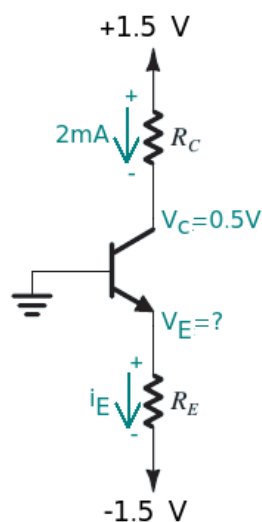
■ As $v_{BE} = v_{BE1} + V_T \ln \frac{i_C}{i_{C1}}$

$$\Rightarrow v_{BE} = 0.8 + (25\text{ mV}) \ln\left(\frac{2\text{ mA}}{1\text{ mA}}\right) = 0.81733\text{ V}$$

- or $v_{BE} = 0.81733 = v_B - v_E = 0 - v_E = -v_E$
- $\Rightarrow v_E = -0.81733V$
- By ohm's law
- $R_E = \frac{v_E - (-1.5)}{i_E}$

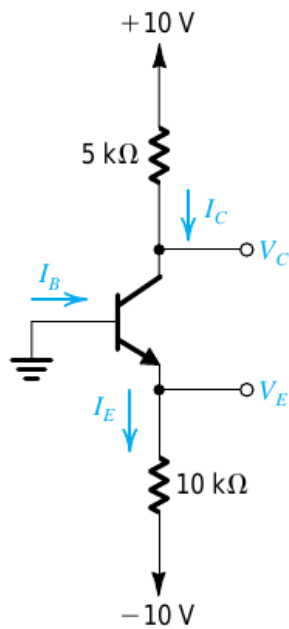


- As $i_E = \frac{i_C}{\alpha} = \frac{2m}{\alpha}$ and $\alpha = \frac{\beta}{\beta+1} = \frac{100}{101} = 0.9901$
- $\Rightarrow i_E = \frac{2m}{\alpha} = \frac{2m}{0.9901} = 2.02mA$
- $\Rightarrow R_E = \frac{v_E - (-1.5)}{i_E} = \frac{-0.81733 - (-1.5)}{2.02m} = \frac{-0.81733 + 1.5}{2.02m} = 337.96\Omega$

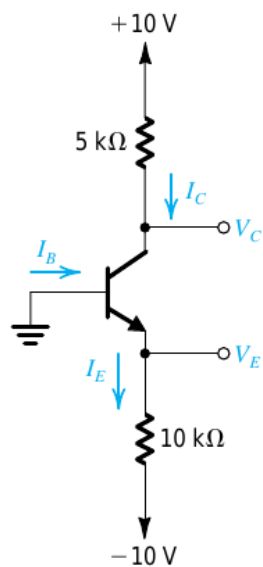


Exercise 6.13

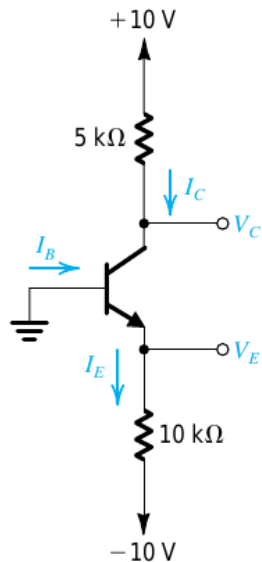
- In the circuit shown in Fig. E6.13, the voltage at the emitter was measured and found to be $-0.7V$. If $\beta = 50$, find I_E , I_B , I_C , and V_C .

**Solution: Exercise 6.13**

- here $v_E = -0.7V$, $\beta = 50$
 - $I_E = ?$, $I_B = ?$, $I_C = ?$, and $V_C = ?$.
 - As $v_E = -0.7V$ and $v_B = 0V$
 - $\Rightarrow v_{BE} = v_B - v_E = 0 - (-0.7) = 0.7V$
 - by ohm's law
 - $i_E = \frac{v_E - (-10)}{10k} = \frac{-0.7 + 10}{10k} = 0.93mA$
 - as $i_E = \frac{i_C}{\alpha}$
 - $\Rightarrow i_C = \alpha i_E$ where $\alpha = \frac{\beta}{\beta + 1} = \frac{50}{51} = 0.98039$
 - $i_C = \alpha i_E = 0.98039 \times 0.93m = 0.91176mA$
 - $i_B = \frac{i_C}{\beta} = \frac{0.91176m}{50} = 18.235\mu A$

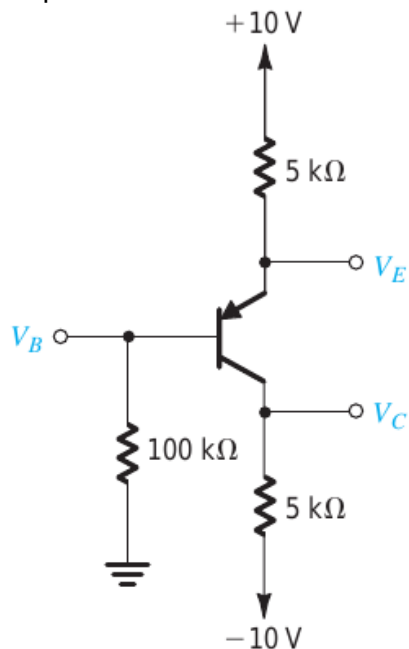


- thus $i_E = 0.93mA$, $i_C = 0.91176mA$, $i_B = 18.235\mu A$
 - by ohm's law
 - $10 - v_C = (5k)i_C = (5k)0.91176m$
 - $10 - v_C = 4.559$
 - or $10 - 4.559 = v_C$
 - $v_C = 5.441V$



Exercise 6.14

- In the circuit shown in Fig. E6.14, measurement indicates V_B to be $+1.0V$ and V_E to be $+1.7V$. What are α and β for this transistor? What voltage V_C do you expect at the collector?



Solution: Exercise 6.14

- here $v_B = 1V$, $v_E = 1.7V$
 - $\alpha = ?$, $\beta = ?$, $v_C = ?$

■ by ohm's law,

$$■ i_B = \frac{v_B}{100k} = \frac{1}{100k} = 10\mu A$$

$$■ \text{ and } i_E = \frac{10 - v_E}{5k} = \frac{10 - 1.7}{5k} = 1.66mA$$

■ As $i_E = i_C + i_B$

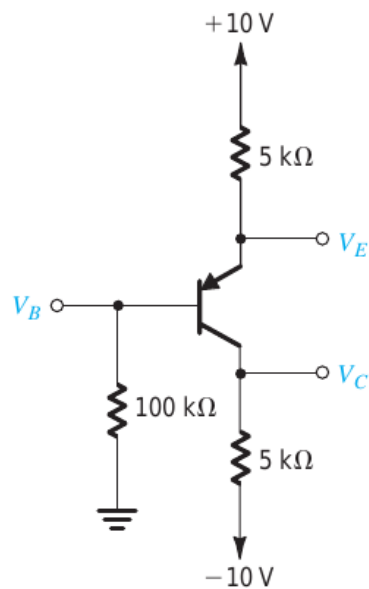
$$■ \Rightarrow i_C = i_E - i_B = 1.66mA - 10\mu A = 1.65mA$$

■ as $i_B = \frac{i_C}{\beta} \Rightarrow \beta = \frac{i_C}{i_B}$

$$■ \Rightarrow \beta = \frac{i_C}{i_B} = \frac{1.65mA}{10\mu A} = 165$$

■ as $i_E = \frac{i_C}{\alpha}$

$$■ \Rightarrow \alpha = \frac{i_C}{i_E} = \frac{1.65mA}{1.66mA} = 0.99398$$

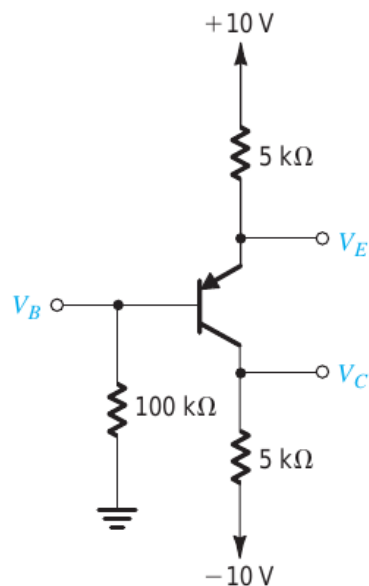


• by ohm's law

$$○ v_C - (-10) = 5k(i_C) = 8.25V$$

$$■ v_C + 10 = 8.25$$

$$■ \text{ or } v_C = 8.25 - 10 = -1.75V$$



BJT: Current-Voltage Characteristics

Graphical Representation of Transistor Characteristics

- for an npn transistor, $i_C = I_S e^{v_{BE}/V_T}$
 - this relation can be represented graphically as
 - also $i_B = \frac{I_S}{\beta} e^{v_{BE}/V_T}$ and $i_E = \frac{I_S}{\alpha} e^{v_{BE}/V_T}$
 - $\Rightarrow i_B - v_{BE}$ and $i_E - v_{BE}$ characteristics are also exponential but with different scale currents
 - $\frac{I_S}{\beta}$ for i_B and $\frac{I_S}{\alpha}$ for i_E

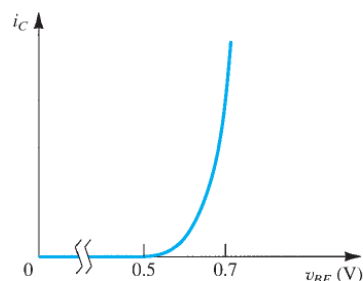


Figure 6.15 The i_C - v_{BE} characteristic for an npn transistor.

- As the constant of the exponential characteristic is $\frac{1}{V_T} = \frac{1}{25m} = 40$
 - $\Rightarrow$ the curve rises very sharply (you can verify this in matlab/octave)
- for v_{BE} smaller than about 0.5V (the cut-in voltage of EBJ), the current is negligibly small
 - Note that for most of the normal range,
 - v_{BE} lies in the range of 0.6V to 0.8V
 - $\Rightarrow$ In performing rapid first order dc calculations,
 - we can utilize the constant-voltage drop model
 - by assuming $v_{BE} \approx 0.7V$

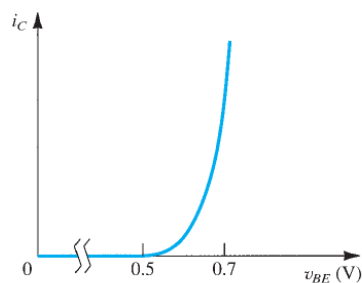
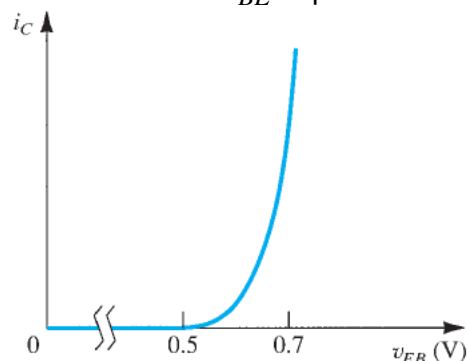


Figure 6.15 The i_C - v_{BE} characteristic for an npn transistor.

- for a pnp transistor, $i_C = I_S e^{v_{EB}/V_T}$
 - $\Rightarrow i_C - v_{EB}$ characteristics will be identical to that of the npn transistor with v_{BE} replaced with v_{EB}



The i_C - v_{EB} characteristic for a pnp transistor.

- Note that for a BJT, the voltage across the emitter-base junction
 - decreases by about 2mV for each rise of $1^\circ C$ in temperature,
 - provided the junction is operating at a constant current

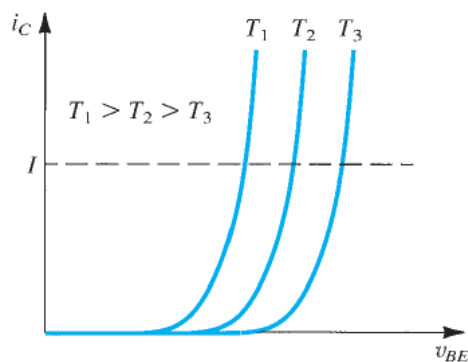


Figure 6.16 Effect of temperature on the i_C - v_{BE} characteristic. At a constant emitter current (broken line), v_{BE} changes by $-2 \text{ mV}/^\circ C$.

Exercise 6.15

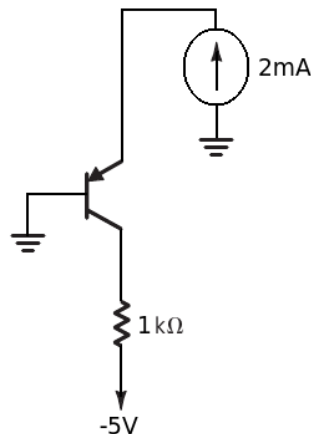
- Consider a pnp transistor with $v_{EB} = 0.7V$ at $i_E = 1mA$. Let the base be grounded, the emitter be fed by a $2mA$ constant-current source, and the collector be connected to a $-5V$ supply through a $1k\Omega$ resistance. If the temperature increases by $30^\circ C$, find the changes in emitter and collector voltages. Neglect the effect of I_{CBO}

Solution: Exercise 6.15

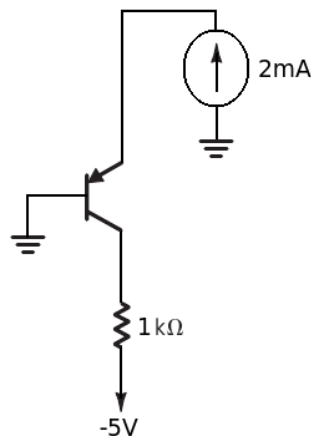
- here pnp, $v_{EB1} = 0.7V$ at $i_{E1} = 1mA$
 - change in v_E and v_C are ?

- if i_{CBO} is neglected
 - $\Rightarrow v_C$ will not change with temperature
 - $\Rightarrow$ change in v_C with temperature = 0
- as v_{EB} decreases by 2mV for every $1^\circ C$ rise in temperature
 - $\Rightarrow v_{EB}$ decreases by 4mV for $2^\circ C$ rise in temperature
 - $\Rightarrow v_{EB}$ decreases by 6mV for $3^\circ C$ rise in temperature

⋮



- $\Rightarrow v_{EB}$ decreases by 60mV for $30^\circ C$ rise in temperature
- v_{EB} decreases by 60mV for $30^\circ C$ rise in temperature
 - as $v_B = 0 \Rightarrow v_{EB} = v_E - v_B = v_E$
 - if v_{EB} decreases by 60mV $\Rightarrow v_E$ will also decrease by 60mV

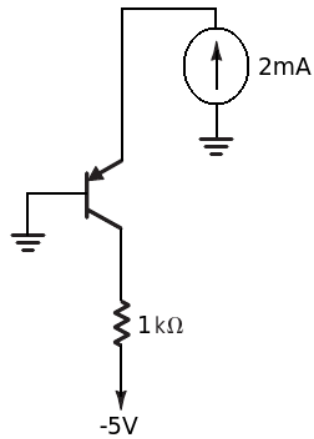


- exact value of v_{EB} can be determined as follows,

- As $i_E = \frac{I_S}{\alpha} e^{v_{EB}/V_T}$
 - $\Rightarrow i_{E1} = \frac{I_S}{\alpha} e^{v_{EB1}/V_T} , i_{E2} = \frac{I_S}{\alpha} e^{v_{EB2}/V_T}$
 - dividing
 - $\frac{i_{E2}}{i_{E1}} = \frac{e^{v_{EB2}/V_T}}{e^{v_{EB1}/V_T}} = e^{(v_{EB2}-v_{EB1})/V_T}$
 - for $v_{EB1} = 0.7V , i_{E1} = 1mA , i_{E2} = i_E = 2mA ,$

$$v_{EB} = v_{EB2} = ?$$

$$\begin{aligned} \blacksquare \frac{i_{E2}}{i_{E1}} &= \frac{2m}{1m} = e^{(v_{EB} - v_{EB1})/V_T} = e^{(v_{EB} - 0.7)/25m} \\ \blacksquare e^{(v_{EB} - 0.7)/25m} &= 2 \\ \blacksquare (v_{EB} - 0.7)/25m &= \ln 2 \\ \blacksquare v_{EB} &= 0.7 + (25m)\ln 2 = 0.71733V \end{aligned}$$



- $v_{EB} = 0.71733V$
 - when temperature increases by $30^\circ C$
 - $\Rightarrow v_{EB}$ decreases by $60mV$
 - i.e. $v_{EB, (at T+30^\circ C)} = 0.71733 - 60m = 0.65733V$
 - $\Rightarrow v_{EB (at T+30^\circ C)} - v_{EB} = 0.65733 - 0.71733 = -60mV$